

CamThink

NeoEyes

NE101

Ultra-Low-Power Event Camera

Wake. Capture. Report.
Then return to sleep.



3+ years*

Theoretical battery life at 10 captures per day in the official Wi-Fi low-power configuration.

Think Beyond Vision

BUILT TO LAST

Capture Only What Matters

NE101 wakes on schedule or sensor input, captures visual evidence, reports it via MQTT, and returns to sleep.

01 Event-Triggered by Design

Wake from a schedule, interval, capture button, PIR sensor or configured GPIO input instead of maintaining a continuous video stream.

02 Battery-First Operation

Deep sleep and firmware-controlled camera, fill-light and storage power domains reduce idle consumption between capture tasks.

03 Image-to-Platform Workflow

Publish JPEG image data with device, battery, time and trigger metadata through MQTT or certificate-enabled MQTTS.

04 Open, Modular Foundation

Interchange camera, communication and sensor modules, then customize the open firmware with ESP-IDF or Arduino.



Periodic Field Observation

Document crops, environments and remote assets without wired power or broadband video.



Meter and Asset Evidence

Capture repeatable images for remote review, record keeping or downstream OCR workflows.



Event-Driven Monitoring

Use PIR or custom sensing to collect evidence only when motion or another condition occurs.

3+
Years*

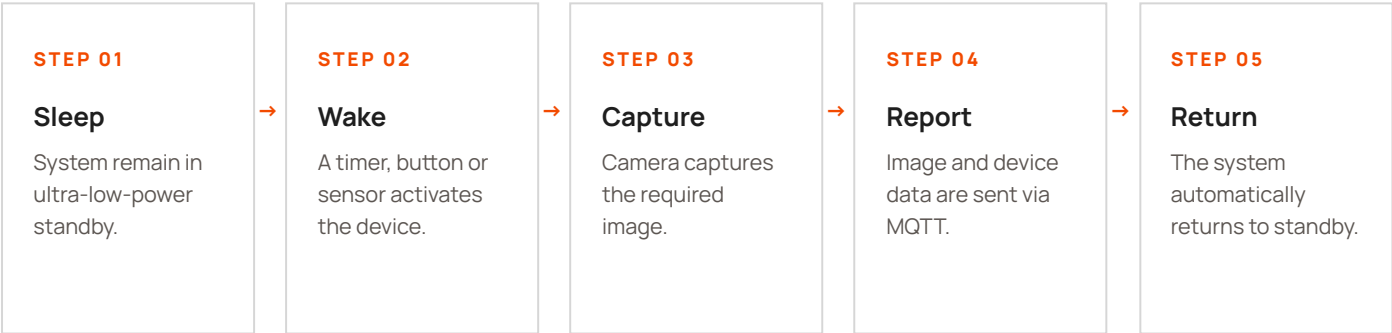
Based on 10 captures per day in the official Wi-Fi low-power configuration.

Low Power by Design

Deep-sleep operation minimizes idle power while preserving scheduled and event-triggered image capture.

One Complete Capture Cycle

LOW-POWER REFERENCE ARCHITECTURE



Scheduled Capture

Configure up to eight daily times across Monday to Sunday for consistent visual records.

Interval Capture

Run one repeating rule based on minute, hour or day intervals for periodic observation.

Sensor-Triggered Capture

Wake from PIR, alarm input or configured GPIO resources, with hardware occupancy considered.

Flexible Upload Timing

Upload immediately after capture or queue data for scheduled transmission at configured times.



Modular Visual Sensing

NE101 combines configurable imaging, wireless and trigger modules in one field-ready camera node.

ESP32-S3 Dual-core, up to 240 MHz	8 MB Octal PSRAM	16 MB Flash Storage	IP67 Protected Enclosure
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STANDARD LENS
OV5640 · 5 MP
2592 × 1944 maximum resolution with 60° or 120° lens and near- or standard-focus options.

OPTIONAL LENS
SC200AI · 1080p
UVC-compliant USB module with an M12 lens mount for alternative fields of view.

WIDE-AREA NETWORK
4G LTE Cat.1
Global and North American module variants for sites without local Wi-Fi or a gateway.

LONG-RANGE NETWORK
Wi-Fi HaLow
IEEE 802.11ah module options for 868 MHz and 915 MHz regional deployments.

Power and Enclosure

4x AA batteries or USB Type-C power; 77 × 77 × 48 mm complete camera.

Expansion Resources

16-pin header provides UART, SPI, GPIO, alarm input and power resources; open firmware supports ESP-IDF or Arduino customization.

Engineering Constraints

Micro-TF conflicts with Cat.1 / HaLow pins; PIR interrupt and alarm input share GPIO2 on the current main board.

Connectivity for Every Site

NE101 supports multiple wireless options, helping each deployment balance coverage, power, infrastructure and operating cost.

STANDARD

2.4 GHz Wi-Fi

Best where an existing access point is available near the camera and image uploads are infrequent.

Ideal for:

Buildings, Labs, Nearby Assets

DIRECT WIDE-AREA

LTE Cat.1

Connect directly through a cellular network when there is no local Wi-Fi or gateway infrastructure.

Ideal for:

Isolated or Mobile Coverage Areas

DISTRIBUTED SITES

Wi-Fi HaLow

Use Sub-1 GHz IEEE 802.11ah for longer-range, low-power links to a compatible gateway.

Ideal for:

Farms, Campuses, Reserves



NE101 Communication Variants

CORE CAMERA PLATFORM REMAINS CONSISTENT

Model	Communication	Region / deployment note
NE101	2.4 GHz Wi-Fi	Standard model for existing Wi-Fi infrastructure
NE101-L01GL	LTE Cat.1	Global frequency-band version excluding North America
NE101-L02NA	LTE Cat.1	North American regional version
NE101-HL00	Wi-Fi HaLow	868 MHz regional version (Global, excluding North America)
NE101-HL01	Wi-Fi HaLow	915 MHz regional version (North America)

Flexible Installation

Wall, desktop, pole, dial and customizable meter brackets support varied viewing angles and mounting surfaces.

Outdoor Operation

IP67 enclosure, tempered-glass lens window and a complete-camera operating range of -20°C to +50°C.

Low Service Burden

Battery operation, local Web setup and event-based MQTT / MQTTS transmission reduce cabling, bandwidth use and routine site visits.

Technical Specifications

ESP32-S3 | 8 MB PSRAM | 16 MB FLASH | OV5640 / SC200AI | WI-FI / CAT.1 / HALOW | IP67

Core Platform

MCU	ESP32-S3
Memory	8 MB PSRAM
Flash	16 MB
Development	Open reference firmware, ESP-IDF and Arduino pathways

Imaging

Standard Camera	OV5640, up to 2592 × 1944
Lens Options	60° / 120° FOV, near or standard focus
Optional Camera	SC200AI USB module, 1920 × 1080
Fill Light	Support

Interfaces and Storage

Storage	Micro-TF card slot
Expansion	16-pin IO header with UART, SPI and GPIO resources
Trigger Interfaces	PIR and alarm input resources

Mechanical

Protection	IP67
Dimensions	77 × 77 × 48 mm
Installation	Wall, desktop, pole, dial and custom brackets
Certification	CE / FCC

Connectivity

Standard Model	NE101 · Wi-Fi
Cellular Models	NE101-L01GL / NE101-L02NA; LTE Cat.1
HaLow Models	NE101-HL00 · 868 MHz; NE101-HL01 · 915 MHz
Data Reporting	MQTT / certificate-enabled MQTTS

Power and Operation

Power Supply	4× AA batteries or USB Type-C
Low-Power Mode	Deep sleep between capture events
Battery Reference	3+ years* at 10 Wi-Fi captures per day
Operating Temperature	-20°C to +50°C
Ambient Humidity	0-90% RH (non-condensing)

Configuration and Control

Capture Rules	Scheduled, interval, button, PIR and alarm-input capture
Upload Rules	Immediate reporting or scheduled upload windows
Local Setup	Device AP and Web configuration interface at 192.168.1.1
Secure Reporting	MQTT or MQTTS with certificate configuration

* Theoretical estimate based on the official Wi-Fi low-power configuration. Actual battery life varies with network conditions, capture frequency, temperature and battery selection.



sales@camthink.ai